34. If $\theta$ is the acute angle between the line $\vec{r} = \vec{a} + t\vec{b}$ and the plane $\vec{r} \cdot \vec{n} = p$, then $\theta = \sin^{-1}\left(\frac{|\vec{b} \cdot \vec{n}|}{|\vec{b}||\vec{n}|}\right)$

35. The perpendicular distance from a point with position vector $\vec{u}$ to the plane $\vec{r} \cdot \vec{n} = p$ is given by $\delta = \frac{|\vec{u} \cdot \vec{n} - p|}{|\vec{n}|}$

36. The perpendicular distance from a point $(x_2, y_1, z_1)$ to the plane $ax + by + cz = p$ is $\delta = \frac{|ax_1 + by_1 + cz_1 - p|}{\sqrt{a^2 + b^2 + c^2}}$.

37. The perpendicular distance from the origin to the plane $ax + by + cz + d = 0$ is given by $\delta = \frac{|d|}{\sqrt{a^2 + b^2 + c^2}}$

38. The distance between two parallel planes $ax + by + cz + d_1 = 0$ and $ax + by + cz + d_2 = 0$ is given by $\frac{|d_1 - d_2|}{\sqrt{a^2 + b^2 + c^2}}$.

39. The vector equation of a plane which passes through the line of intersection of the planes $\vec{r} \cdot \vec{n}_1 = d_1$ and $\vec{r} \cdot \vec{n}_2 = d_2$ is given by $(\vec{r} \cdot \vec{n}_1 - d_1) + \lambda(\vec{r} \cdot \vec{n}_2 - d_2) = 0$, where $\lambda \in \mathbb{R}$ is an.

40. The equation of a plane passing through the line of intersection of the planes $a_1x + b_1y + c_1z = d_1$ and $a_2x + b_2y + c_2z = d_2$ is given by
$$(a_1x + b_1y + c_1z - d_1) + \lambda(a_2x + b_2y + c_2z - d_2) = 0$$

41. The position vector of the point of intersection of the line $\vec{r} = \vec{a} + t\vec{b}$ and the plane $\vec{r} \cdot \vec{n} = p$ is $\vec{u} = \vec{a} + \left(\frac{p - (\vec{a} \cdot \vec{n})}{\vec{b} \cdot \vec{n}}\right)\vec{b}$, where $\vec{b} \cdot \vec{n} \neq \vec{0}$.

42. If $\vec{v}$ is the position vector of the image of $\vec{u}$ in the plane $\vec{r} \cdot \vec{n} = p$, then $\vec{v} = \vec{u} + \frac{2\left[p - (\vec{u} \cdot \vec{n})\right]}{|n|^2}\vec{n}$ .

## ICT CORNER

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